Sahaj Bhadja

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Summary

AI Engineer with 2+ years building production **LLM** systems: fault-tolerant orchestration, multi-model routing with fallbacks, **RAG** retrieval, and evaluation, plus **security** research in malware detection and authentication, in **Python** and TypeScript.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **LLM orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Built data pipelines feeding AI workflows across PostgreSQL, SQL Server, and cloud storage with validation and reconciliation, holding 99%+ production **reliability** and securing tenant data with row-level security.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a production **multi-step LLM** system chaining rubric generation, OCR extraction, grading, and reasoning across GPT-4o, Claude, and Qwen-VL, serving 10K+ daily requests for 150+ instructors at 90%+ consistency.
- Designed a provider-agnostic model layer with dynamic switching, timeout-aware **fallback**, and per-call token and cost monitoring, keeping LLM output reliable while cutting average feedback turnaround 60% for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized large-scale **API integrations** with object-oriented Singleton and Factory designs, caching, and pagination, improving response times 30% and cutting technical debt 20% across a cross-platform microservice platform.
- Automated 20+ unit and integration **test** suites with PyTest and Mockito in nightly Jenkins CI/CD across three operating systems, and led an ETL redesign boosting pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and Node.js microservices with React front ends for authentication, role-based access, billing, and notifications, syncing 2M+ records across multiple locations in real time via WebSockets.
- Integrated Snyk and OWASP Dependency-Check **security scanning** into Jenkins CI/CD with unit, integration, and end-to-end tests, reaching 95%+ build validation and tracking production errors and regressions in Sentry.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Built a **malicious URL detection** pipeline over 650K+ URLs, engineering 21 behavioral indicators such as IP presence, URL length, and suspicious keywords to classify phishing and malware.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN models, reaching 96%+ **threat detection** accuracy and reporting actionable insights into phishing and malware trends.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java
AI & Agents: LLM Orchestration, Multi-Step LLM Pipelines, Multi-Model Routing, RAG, Embeddings, FAISS, Pinecone, Claude, GPT-4o
ML & Security: PyTorch, Transformers, Feature Engineering, Malware Detection, Phishing Detection, Snyk, OWASP, JWT, RBAC
Backend & Frontend: FastAPI, Flask, Node.js, REST APIs, React, Next.js, Background Workers, Microservices
Data & Infra: PostgreSQL, SQL Server, MongoDB, AWS, Azure, Docker, CI/CD, Sentry, Monitoring
Practices: Claude Code, Code Reviews, Unit Testing, PyTest

Projects

Vulnerability Triage Agent | *Python, Claude, FastAPI, pgvector, PostgreSQL, Semgrep, React*

- Designed a tool-calling **security agent** that enriches Semgrep findings with retrieved past tickets and incidents, judges exploitability from code context, drafts fixes, and is scored on a labeled CVE **eval** set.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Embeddings, PostgreSQL*

- Built a **RAG** system over 50K+ documents with transformer embeddings in FAISS and Pinecone and an incremental ingestion pipeline, reaching 95% query accuracy and cutting retrieval latency 40%.

Comprehensive Analysis of Blockchain for User Authentication Paper | *Authentication, RBAC, Threat Modeling*

- Co-authored an IEEE paper evaluating blockchain **authentication** for users and IoT devices, threat-modeling identity theft, replay, man-in-the-middle, and phishing attacks against centralized identity stores and smart-contract-based verification schemes.